

GIModel.jl: An Inspectable Quark Model for Computation and Learning

GIModel reproduction project

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Abstract

GIModel.jl makes the Godfrey–Isgur relativized quark model available as an inspectable Julia computation. This report describes the software, its reproduction evidence, and the learning material released alongside it. The original 1985 paper supplies the physics; a companion theory course and Pluto notebooks let readers work through that physics, change inputs, and follow a result to its operators, wavefunctions, and numerical convergence. The package provides a computational foundation for questions about meson masses, mixing, and decays, including questions posed through a language model interface. The same numerical objects support interactive flavor comparisons, scientific visualizations, and future extensions of the model. A separate comparison package records the provenance and residuals of the reproduction, including unresolved photon-decay coverage. An appendix traces the agent-assisted development from a finite-difference reference calculation back to the paper’s harmonic-oscillator method, documenting the failures and corrections that made the underlying physics clearer.

1 A computational reference for hadron physics

Quantum mechanics provides a concrete foundation for learning hadron physics: Hamiltonians produce spectra, wavefunctions determine matrix elements, and mixing changes both the identities of states and their observable transitions. The relativized quark model of Godfrey and Isgur (Godfrey and Isgur 1985) brings these ideas together across light and heavy mesons. Working through it connects familiar quantum-mechanical operations to an extensive phenomenological calculation.

GIModel.jl makes that calculation available for reuse and inspection. A reader can ask for a meson spectrum, examine a particular state, resolve its mixed components, and follow the calculation down to the radial operators. The scientific contribution of this project is the recovery and organization of that executable knowledge: equations, numerical conventions, reference data, and checks that can be used again by someone other than their implementer.

This report serves as a reference for citing GIModel.jl and understanding what it supplies. The original paper remains the reference for the model itself. The accompanying theory sheets and computational notebooks provide the detailed learning route. Here we explain why the software is useful,

how its answers can be inspected, what supports the reproduction, and what becomes possible when the technical ingredients are openly available.

There are four complementary deliverables. **GIModel** owns the physics computation. **GIPaper** owns the comparison with the original paper, including digitized inputs and reference assignments. **The learning track** supplies problems, worked solutions, and computational exploration. **This report** connects those resources and records their scope. Together they fill a practical gap between a published model and a reusable tool in the open-source research ecosystem. The value of filling that gap does not depend on a claim to be the first quark-model program: it lies in making this particular calculation available with its assumptions and evidence attached.

2 From a question to a numerical answer

2.1 Masses, mixing, and decay

A useful physics interface starts with questions. What contributes to a meson mass? Why do two states with the same quantum numbers mix? Why can a small change in mixing produce a large change in a decay amplitude? GIModel supplies objects with which these questions can be made precise.

In the model, a rest-frame mass is an eigenvalue of the Hamiltonian, whose kinetic contribution is

$$H_0 = \sqrt{p^2 + m_1^2} + \sqrt{p^2 + m_2^2}.$$

Confinement and spin-dependent interactions determine the rest of the operator. The solver produces both eigenvalues and wavefunctions. When the allowed configurations mix, a physical state takes the form

$$|\Psi\rangle = \sum_a c_a |\psi_a\rangle,$$

with signed coefficients and explicit flavor content. A transition then involves a coherent sum,

$$\langle\Psi_f|O|\Psi_i\rangle = \sum_{a,b} c_{f,a}^* c_{i,b} \langle\psi_{f,a}|O|\psi_{i,b}\rangle.$$

This expression explains why a mixing probability alone cannot predict a decay: relative phases can produce enhancement or cancellation. A width additionally requires

the specified decay operator, normalization, and kinematics; it is not determined by the mass spectrum alone.

The package exposes the stages needed to investigate such an answer. A spectrum contains identifiable states; their physical components lead to solver-native waves; wave operations lead to radial matrix elements. The comparison layer then relates the result to a printed observable. The distinction between an input, a computed quantity, and a reference value remains visible throughout.

2.2 A language model interface grounded in computation

This organization supports a conversational way of using the model. A language model can translate a physics question into a package calculation, help explain the returned objects, and guide the reader to the methods involved. For example, “why does this mixing suppress the transition?” can lead to the individual signed contributions to an amplitude and a repeat calculation with a controlled change in the mixing or Hamiltonian.

Reliability comes from that inspectable calculation. A useful numerical answer records the chosen model parameters, state labels, solver settings, and convergence information, then makes the intermediate quantities accessible. The language model provides navigation and explanation; the numerical evidence comes from executing the package. This is an intended use of the existing computational interface, rather than a claim that an independently validated conversational product has already been released.

The broader teaching ambition includes discussing mesons and baryons in the same quantum-mechanical language. The implemented GIModel calculation described here is for mesons. A quantitative baryon tool would require its own three-body dynamics, state construction, and validation. The present package provides a worked example of how to make such a discussion computationally accountable.

3 Learning by working through the model

3.1 Theory and computation as companion materials

The learning material is intended to train physicists who can work through a model in detail, connecting a paper reference to an actual calculation. The current theory course contains nine sheets, progressing through radial quantum mechanics, angular momentum and meson quantum numbers, variational and semirelativistic methods, confinement and running coupling, relativization and smearing, spin fine structure, mixing, decay observables, and the complete spectrum algorithm. Problems have worked solutions and short concept checks. A computational-discovery sheet begins the bridge to numerical work; the extended hands-on course is a development direction.

This division gives each resource a clear role. The paper presents the original physical model and its reasoning. The theory course slows down the derivations and supplies exercises. The notebooks make assumptions adjustable and the consequences visible. The report explains the resource as a whole without repeating the full theory course.

3.2 Pluto and live documentation

Pluto provides a useful set of teaching primitives: executable cells, reactive dependencies, mathematical prose, controls, and immediate displays of results. A student can change a mass or numerical setting, observe the dependent output, and investigate the method responsible. Live documentation and linked docstrings make this an entry point into the package rather than an isolated demonstration. The intended reading path is short: identify a meson, compute a spectrum, select a state, inspect its components, and examine the relevant wave or matrix element.

The documentation graph explicitly links these public operations. A static link audit checks their connectivity; it establishes a navigable source-level structure, while the notebook remains the place where a reader experiences that structure. Discoverability is part of the teaching design: an explanation should lead naturally to the next numerical question.

The adaptive harmonic-oscillator notebook makes convergence itself a lesson. It follows the oscillator scale and basis refinements, displays the changes in energies, and finishes with a comprehension quiz. Students can see why an apparently stable curve needs a numerical criterion and why energy convergence alone does not establish every wave-sensitive observable. The computation therefore teaches both the model and the standards needed to trust its output.

4 Reproduction as inspectable evidence

4.1 Recovering the calculation in depth

The reproduction program reaches through the spectrum to mixing, strong decays, photon transitions, and annihilation observables, together with the appendix machinery behind them. Its purpose is to recover the paper at the level of numerical operations and conventions. A matching mass is one check; recovering the wavefunction and using it consistently in another observable is a stronger one.

The paper’s spectrum procedure uses a harmonic-oscillator (HO) basis. The current native implementation first diagonalizes the full relativized Hamiltonian within fixed angular-momentum and spin sectors. It then forms the allowed tensor and antisymmetric spin-orbit mixing blocks from those solutions. Self-conjugate isoscalar channels receive the annihilation and flavor-mixing stage. Observables consume the resulting physical composition. A separate

finite-difference (FD) implementation supplies an independent numerical comparator.

The HO calculation evaluates position- and momentum-dependent factors in their native representations, following Appendix A. The oscillator scale is optimized and the basis enlarged until every requested energy passes the stated convergence test. The checked convergence report uses a criterion of 0.1 MeV on two consecutive basis refinements; wave-overlap diagnostics and wave-sensitive audits supply additional evidence. This numerical tolerance is a solver criterion, not an estimate of the physical accuracy of the model.

4.2 Provenance and remaining discrepancies

GIPaper keeps a manifest of paper units with links to code, tests, reference sources, and residual reports. This is the detailed record of what has been computed and compared. Reference data retain their origin in printed tables or digitized figures. Parameter provenance also matters: taking the paper’s inputs, repeating a fit performed in the paper, and calibrating a diagnostic control to reference targets are different operations. Reports label these choices so that a control cannot silently become evidence for an unfitted prediction.

As one compact example, the photon audit reports $\mu/\mu_N = +0.688$ for $\rho \rightarrow \pi\gamma$, compared with the printed $+0.69$, and -1.200 for $K^{*0} \rightarrow K^0\gamma$, compared with -1.20 . These are values from the checked-in September 7, 2026 audit, not a fresh calculation for this manuscript. The ρ row was used by the original paper to fit a relativization exponent, so its agreement checks the reconstructed pipeline rather than supplying an independent prediction. Other transitions probe different flavor weights, wavefunctions, and cancellations.

The release aim is full-depth reproduction of the original results. The current repository audit still identifies complete accounting of Table VI photon-decay rows and characterization of its remaining residuals as unfinished. Some mixing and cancellation-sensitive quantities also retain visible discrepancies. Accordingly, the present report does not claim exact agreement for every printed number or completed whole-paper coverage. Those boundaries belong beside the reproduction evidence and should be updated when the corresponding audit closes. The native three-stage spectrum implementation and the independent FD comparator are already documented separately from that remaining task.

5 Visualizing the physics

5.1 Connecting flavor sectors

The heavy-quark notebook demonstrates a particularly useful consequence of an executable model. The constituent heavy-quark mass can be varied continuously while the

shared interaction parameters remain fixed. With one light or strange partner, this connects the model’s D and B sectors or its D_s and B_s sectors. Varying both constituent masses connects charmonium and bottomonium. The existing notebook provides these three choices, live spectrum views, and optional recorded sweeps.

Subtracting the constituent rest masses helps separate the change in binding and excitation structure from the rise in total mass. The reader can follow hyperfine splittings, level ordering, and the response of the light degrees of freedom as the heavy mass changes. Heavy masses act here as external parameters of a constituent model with a shared phenomenological interaction. This is a useful way to explore the model; it does not assert that its constituents are literally outside the QCD vacuum. The interpolation is also a family of model calculations, not a physical decay from a bottom meson to a charmed meson.

5.2 Meson structure and animations

The same access to wavefunctions supports a further teaching direction: visualizing meson structure. Radial probability densities, nodes, characteristic radii, angular distributions, and mixed-state components can be displayed alongside the spectrum. An animation could follow how these quantities change with constituent mass or an interaction parameter, recomputing the state at each step.

Such a visualization should identify what it displays. For a normalized reduced radial wave, $|u(r)|^2 dr$ is a radial probability, whereas the three-dimensional density includes the radial and angular factors. Mixed states require their coherent composition and the appropriate spin or flavor sums. These distinctions make a scientifically interpretable structure picture possible. They also supply exercises: a student can connect the number of nodes to a radial excitation or relate short-distance structure to an annihilation matrix element. The technical ingredients are available; a curated meson structure animation remains a proposed addition.

6 A foundation for further work

An open implementation lets new work start from a checked calculation. Researchers can reuse the state construction, matrix elements, wave operations, and convergence machinery, while the comparison package preserves a baseline against which a change can be understood. Teaching notebooks and scientific visualizations are uses of the same computational objects, so improvements in the core can propagate to both.

Post-GI developments provide natural physics directions. Examples include screened interactions (Li and Chao 2009) and explicit coupled-channel treatments (Barnes and Swanson 2008; Ortega et al. 2010). Each changes the physical model and requires its own validation. Another concrete extension is to apply the paper’s quark-emission operator directly to the calculated physical wavefunctions, going

beyond the single-oscillator-scale approximation used for its numerical strong-decay tables. This would expose the role of radial nodes, state-dependent scales, and mixing in those amplitudes.

Julia also makes parameter-sensitive computation a promising direction. For an isolated normalized eigenstate, the Hellmann–Feynman relation provides an explicit starting point,

$$\frac{\partial E_n}{\partial \lambda} = \left\langle \psi_n \left| \frac{\partial H}{\partial \lambda} \right| \psi_n \right\rangle.$$

Such derivatives could support sensitivity studies and fitting. The repository already contains preparation of numerical types and FD and fixed-scale HO derivative probes. End-to-end automatic differentiation is future work: adaptive basis selection, crossings, and derivatives of mixed eigenvectors need careful treatment. The opportunity rests on a transparent Julia implementation, with the derivative claim subject to checks just as the original computation is.

The lasting purpose of the release is to let readers do physics with the model: ask a question, obtain a numerical answer, inspect its origin, and change the calculation knowingly. The course, notebooks, and documentation make that process accessible to a new generation of physicists.

A Development as a record of learning

A.1 From a finite-difference foothold back to 1985

The development took a useful historical detour. In 1985 the HO basis provided the paper’s practical route through an operator that depends on both position and momentum. Oscillator functions have tractable representations in both spaces; Appendix A inserts intermediate states to assemble products of the corresponding matrix elements. The numerical representation was part of what made that calculation workable.

For this project, an FD radial calculation was the easier initial foothold. Grid values, operator matrices, and contribution plots gave a direct way to inspect a first reference implementation. Once it worked, the independent result supported the return to native HO matrix elements and then to the paper’s complete order of diagonalization and mixing. The sequence was thus FD to HO, with repeated comparisons along the way. This is a lesson from this project’s development, rather than a general ranking of the two methods.

The early comparison also concealed an important distinction: representing a wave in an HO expansion does not by itself reproduce the paper’s algorithm. Native operator assembly, full fixed-sector diagonalization, and the later mixing stages all had to be recovered. The eventual result retained FD as an independent comparator and made the paper’s HO route self-contained.

A.2 Corrections that taught physics

Normalization and relative phases. A diagonal Rayleigh quotient is unchanged when its input vector is rescaled; a bare off-diagonal overlap is not. Comparing sampled FD waves and reconstructed HO waves therefore required a common physical normalization. Otherwise seemingly matching diagonal results could coexist with bad mixing matrix elements. The normalization work is recorded in commit [a987862](#). The HO phase-convention correction in [9ff437b](#) illustrates a related point: an individual eigenvector has an arbitrary phase, but every operator and mixed-state composition must transform consistently with it. These are physical consistency lessons revealed by implementation failures.

A converged approximation can still be the wrong calculation. An unavailable nonperturbative solve had previously fallen back to a first-order shift. Commit [eef789b](#) made that case fail explicitly. The later fixed-sector work ([b9d3474](#)) put the full diagonal spin interactions into the Hamiltonian before solving it. Recovering masses alone was insufficient: spin-distorted wavefunctions had to reach decay and annihilation observables as well. The completion in [78a3d76](#) also removed extra spin-strength bridge factors after correcting the literal Appendix-A coefficients.

Native matrices and observable convergence. Commit [8655d68](#) removed the mesh from the HO Appendix-A assembly. The adaptive convergence work in [39ee9c8](#) then made basis refinement an explicit production requirement. Convergence of energies still had to be complemented by tests of origin-sensitive functionals and final mixed-state observables. The independent FD certification in [bdd65d8](#) supplied a second numerical route with its own spacing and domain checks. Agreement became evidence about the calculation only after the representations and approximations were identified.

Verification infrastructure can hide a mistake. Commit [959fdb0](#) corrected a scorecard that reported the previous verification run; [2fd3fd7](#) restored FD/HO audits that had fallen outside the gate. Such failures matter scientifically: a correct-looking report may not describe the code being inspected. Keeping the reference comparison and its generation path explicit is part of reproducibility.

A.3 Agent-assisted coding and the evidence trail

The git history records extraction, implementation, autonomous iterations, physics audits, repairs, and teaching material. Agents helped carry out this work, while the source paper, numerical checks, and human review supplied the criteria for accepting it. The useful story includes the false starts: plausible code, successful local tests, or a reassuring residual could leave a missing physical operation undiscovered.

The examples above show why the corrective loop mattered.

A discrepancy prompted a question about normalization, operator staging, or provenance; that question led back to the paper or an independent calculation; the answer changed both code and explanation. These episodes can teach quantum mechanics alongside software practice. They are case studies from one repository, not a controlled measurement of agent productivity or reliability.

The commit identifiers here are navigation anchors into that record. Detailed changes and their audits are better evidence than a commit title alone. A companion website story could pair each episode with its original discrepancy, the inspected equation, the fix, and an interactive reconstruction. This appendix provides the narrative spine for that story; the report’s main argument remains the usefulness of the resulting scientific and educational resource.

B Repository reading guide

The following paths locate the supporting material within the release.

- **Learning:** `LearningTrack/README.md` and `examples/README.md` introduce the theory sheets and computational notebooks.
- **Discoverability:** `docs/discoverability_graph.md` records the linked public API; `docs/formula_map.md` connects implementation to paper equations.
- **Reproduction:** `docs/paper_manifest/` and `docs/paper_dashboard.qmd` provide the per-unit inventory. `docs/reproduction_audit.md` records the cross-cutting interpretation and remaining scope.
- **Numerical evidence:** `GIPaper/docs/residual_reports/` contains the HO convergence, FD comparator, spectrum, mixing, and decay audits. The numerical example in this report comes from `table_vi_photon_decays.md` there.
- **Development:** `docs/original_1985_algorithm_audit.md`, `docs/incident_followup_audit.md`, and the git history document the recovery of the paper’s algorithm and the associated corrections.
- **Differentiation:** `diff_support/implementation_status.md` distinguishes completed preparation and probes from proposed derivative capabilities.

References

- Barnes, Ted, and Eric S. Swanson. 2008. “Hadron Loops: General Theorems and Application to Charmonium.” *Phys. Rev. C* 77: 055206. <https://doi.org/10.1103/PhysRevC.77.055206>.
- Godfrey, Stephen, and Nathan Isgur. 1985. “Mesons in a Relativized Quark Model with Chromodynamics.” *Phys. Rev. D* 32: 189–231. <https://doi.org/10.1103/PhysRevD.32.189>.
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